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Highly efficient near-infrared to visible up-conversion process in $NaYF_4$: Er^{3+}, Yb^{3+}

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Abstract

Up-conversion (UC) entails the addition of two (or more) photons to generate one of higher energy. This process is very interesting, both from a fundamental point of view as well as for (future) applications. Here, dependence of the emission spectra on excitation power and measurement temperature are reported for the very efficient near-infrared to visible photon UC material $NaYF_4$: 2% Er^{3+}, 18% Yb^{3+}. In the high-power limit for continuous-wave laser excitation, roughly 50% of all near-infrared photons absorbed by the material are up-converted and emitted in the visible spectral range. The excitation spectrum shows a $39\,cm^{-1}$ energy gap between the two lowest crystal-field components $|0\rangle$ and $|1\rangle$ of the Yb^{3+} $^2F_{5/2}$ multiplet, which sensitizes the UC emission. There is very efficient energy transfer to the lowest energy crystal-field component of the Er^{3+} $^4I_{11/2}$ state from $^2F_{5/2}|1\rangle$, resulting in an activation energy for all Er^{3+} related up- and down-conversion emissions that scales with the number of excitation photons required multiplied by the $39\,cm^{-1}$ energy gap.
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1. Introduction

Fascinating and novel phenomena are attributed to photon up-conversion (UC) in inorganic materials [1]. Applications such as lasers [2], next generation imaging devices [3] and near-infrared (NIR) quantum counting devices [4] have been demonstrated. As high-power NIR lasers are affordable and easily obtainable, efficient NIR to visible UC is expected to have significant technological impact. Besides future applications, a great interest also exists in the fundamental physics of efficient UC materials due to the interesting interplay between linear and non-linear optical

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excitation processes, resulting in complex dependencies on the excitation power and non-trivial luminescence transients [5]. Studying these energy transfer processes will further support the understanding of UC mechanisms in general and can lead to better tailored UC materials.

Since the early 1970s, $NaYF_4:Er^{3+},Yb^{3+}$ is known as one of the most efficient NIR to visible UC materials [6–10], but no systematic study of the physical properties of this material has been undertaken as of yet. The most important omissions in the literature are the total spectral distribution (including all NIR emissions) as well as the power dependence and the temperature dependence of the emission bands under NIR excitation. So far, these have not been investigated in detail for any NIR to visible UC material. In this paper, a study of these properties of $NaYF_4:Er^{3+},Yb^{3+}$ will be presented as well as a proof of the highly efficient UC mechanism involved. From the experimental data a determination of the absolute efficiency as well as the fundamental (highly efficient) excitation mechanism for near-infrared to visible UC in $NaYF_4:Er^{3+},Yb^{3+}$ is presented. Our investigation proves that this material is a very efficient NIR to visible up-converter due to the near-perfect resonance between a specific crystal-field level of Yb^{3+} and one of Er^{3+}. As a result, the UC can take place through direct (resonant) energy transfer from Yb^{3+} to Er^{3+} and no phonon-mediated step is required. These results are pivotal for understanding the properties of this material as well as for its application in a device. From our data, it is also clear that the red to green emission ratio changes with excitation power, which will be important when functionalizing this material in a device. This change in red to green emission ratio is indicative of a change in excitation mechanism, which in itself is interesting from a fundamental point of view.

2. Experimental

Our group recently published the optimized synthesis of high-quality $NaYF_4:Er^{3+},Yb^{3+}$ powder [11]. This study indicated that the material should be in the hexagonal phase with Er^{3+} and Yb^{3+} concentrations of 2% and 18%, respectively, to obtain the highest UC intensity. The route described in Ref. [11] was used to fabricate the material used in this study.

The powders were excited with a multimode, standing wave Ti:sapphire laser (Spectra Physics 3900S), pumped by an argon-ion laser in all-lines mode (Spectra Physics 204515/4S) or the second harmonic of a Nd:YVO_4 laser (Spectra Physics Millennia CS-FRU). The wavelength control of the Ti:sapphire laser was achieved by an inchworm-driven (Burleigh PZ-501) birefringent filter and a wavemeter (Burleigh WA2100). Sample cooling was achieved using a quartz helium gas flow tube. The sample luminescence was dispersed by a 0.85 m double monochromator (Spex 1402) with gratings blazed at 500 nm (1200 grooves/mm). The signal was detected with a cooled photomultiplier tube (Hamamatsu R3310) and a photon counting system (Stanford Research SR400). Device control and data acquisition were done by a personal computer. The luminescence spectra were corrected for the instrument response and are displayed as a photon flux per constant energy interval [12]. To measure the power dependence, the beam was attenuated with a series of neutral density filters (Balzers). The luminescence background of the (unfocussed) Ti:sapphire laser was removed using a double prism.

3. Results and discussion

Fig. 1 shows room temperature emission spectra of $NaYF_4:Er^{3+},Yb^{3+}$ for three different excitation powers, chosen to be in the low-, intermediate- and high-power regimes, respectively. For excitation powers $\gtrsim 400$ mW no further change in relative emission intensities is observed [13]. For each spectrum, the area under emission band i corresponds to the number of photons emitted in that band [12], and is denoted by Φ_i. As a result, $f_i \equiv \Phi_i / \sum_{\forall j} \Phi_j$ represents the fraction of all photons emitted in band i. The assignment of the electronic transition responsible for each band is given in Table 1 together with the fraction of all photons emitted in each of the six emission bands, for the

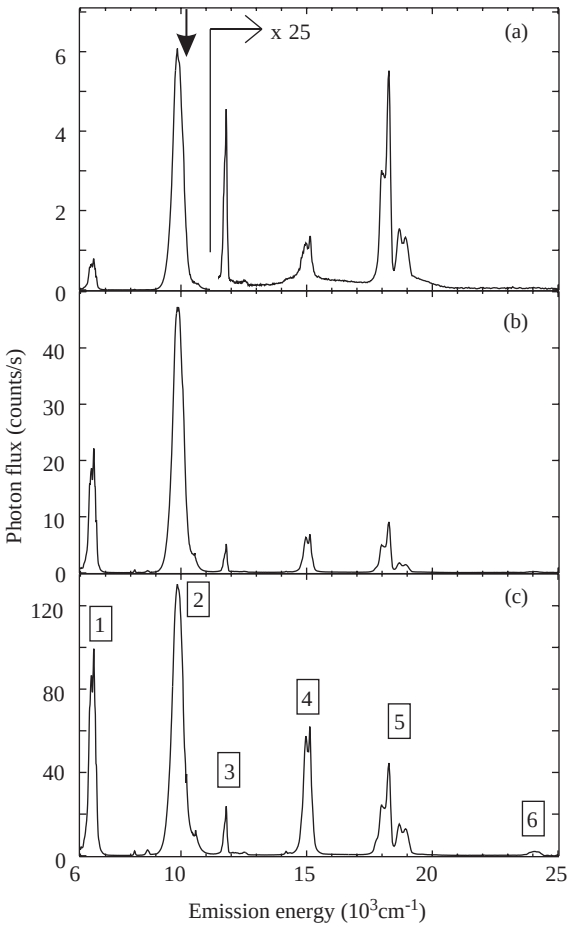


Fig. 1. Emission spectra of $NaYF_4$:Er^{3+},Yb^{3+} excited at 10 238 cm^{-1} (indicated by the arrow, laser-line removed manually) with (a) 10, (b) 100, (c) 600 mW. Recorded at room temperature and relative intensities can be compared. Relative photon fractions and assignment of all emissions, as labeled in Fig. 1(c), are shown in Table 1.

different excitation powers used. Due to the overlap of two emissions around 10 000 cm^{-1} and two around 18 200 cm^{-1}, no attempt is made to resolve the separate emissions and only one value is shown for the total band. Further emission measurements have been done in the spectral region down to 3300 and up to 35 000 cm^{-1}, but no significant emissions were observed outside the region shown in Fig. 1.

The excitation spectrum of the 18 200 cm^{-1} UC emission, shown in Fig. 2(a), was recorded at 5 K to minimize temperature-induced broadening. This spectrum is typical for Yb^{3+} $^2F_{7/2} \rightarrow {}^2F_{5/2}$ absorption. The excitation spectra of the other emission bands are similar. At increased temperature the lines broaden, but the Yb^{3+} characteristics remain pronounced. From Fig. 2(a) the two lowest energy crystal field components within the $^2F_{5/2}$ multiplet are identified: 10 199 and 10 238 cm^{-1}, denoted as $^2F_{5/2}|0\rangle$ and $^2F_{5/2}|1\rangle$, respectively. The broader excitation bands at higher energy are attributed to $^2F_{5/2}|2\rangle$ and phonon-assisted sidebands. Fig. 2(b) shows the excitation spectrum of the same emission (recorded at 12 K) in a $NaYF_4$:10% Er^{3+} sample. The lines correspond to electronic and vibronic excitations of the Er^{3+} $^4I_{11/2}$ multiplet. The lowest crystal field level of this multiplet, denoted by $^4I_{11/2}|0\rangle$, is at 10 244 cm^{-1}. Note that this is at higher energy than the $^2F_{5/2}|0\rangle$ level of Yb^{3+}, but in near-perfect resonance with $^2F_{5/2}|1\rangle$. At room temperature the excitation lines are homogeneously broadened and there is optimal spectral overlap for $^2F_{5/2}|1\rangle \rightarrow {}^4I_{11/2}|0\rangle$ energy transfer. From comparing Figs. 2(a) (circled part) and (b) it is evident that even at 5 K more than 95% of the UC emission in the co-doped sample is Yb^{3+} sensitized.

Let p_i denote the number of NIR excitation photons required to induce emission in band i (the values of p_i are indicated in Table 1) and Φ_i the number of photons emitted in that band. Then $T \equiv \sum_{\forall j} p_j \Phi_j$ is the (minimum) number of NIR excitations required to induce the emission spectrum. It was found that T is independent of temperature when corrected for the temperature-dependence of the Yb^{3+} excitation cross-section [14]. This implies that between 5 and 300 K no significant non-radiative multiphonon relaxation to the Er^{3+} or Yb^{3+} ground state occurs, because such non-radiative processes would depend on the sample temperature. Furthermore, the luminescence lifetimes of the Yb^{3+} $^2F_{5/2}$ and Er^{3+} $^4I_{13/2}$ states were measured under direct excitation and both were found to be in the millisecond-range and effectively independent of the sample temperature. Therefore, each photon absorbed must contribute to the emission of a photon and this

Table 1
Fraction of all photons emitted in each band (denoted by f_i) as well as the fraction of excitation photons absorbed that are subsequently used to excite each of the emission bands (denoted by R_i and defined in Eq. (1)) for the six significant emission bands in $NaYF_4:Er^{3+},Yb^{3+}$ under 10 238 cm^{-1} excitation at room temperature (data shown in Fig. 1). The different excitation powers are indicated. Also shown are the assignments of the emission bands and the (minimum) number of excitation photons needed to excite these emission bands (denoted by p_i)

i	Energy of emission band maximum (cm^{-1})	Electronic transition involved	p_i	f_i			R_i		
				10 mW	100 mW	600 mW	10 mW	100 mW	600 mW
1	6500	$Er^{3+}:{}^4I_{13/2} \rightarrow {}^4I_{15/2}$	1	3.0%	13%	18%	2.9%	10.8%	13.4%
2	10 000	$Yb^{3+}:{}^2F_{5/2} \rightarrow {}^2F_{7/2}$ $Er^{3+}:{}^4I_{11/2} \rightarrow {}^4I_{15/2}$	1	93%	68%	48%	90.3%	56.8%	35.9%
3	11 800	$Er^{3+}:{}^4S_{3/2} \rightarrow {}^4I_{13/2}$	2	0.6%	1.7%	2.1%	1.2%	2.8%	3.1%
4	15 000	$Er^{3+}:{}^4F_{9/2} \rightarrow {}^4I_{15/2}$	2	0.5%	7.9%	16%	1.0%	13.2%	24%
5	18 200 18 800	$Er^{3+}:{}^4S_{3/2} \rightarrow {}^4I_{15/2}$ $Er^{3+}:{}^2H_{11/2} \rightarrow {}^4I_{15/2}$	2	2.6%	9.5%	15%	4.9%	15.8%	22.5%
6	24 000	$Er^{3+}:{}^2H_{9/2} \rightarrow {}^4I_{15/2}$	3	0.01%	0.2%	0.5%	0.03%	0.50%	1.1%

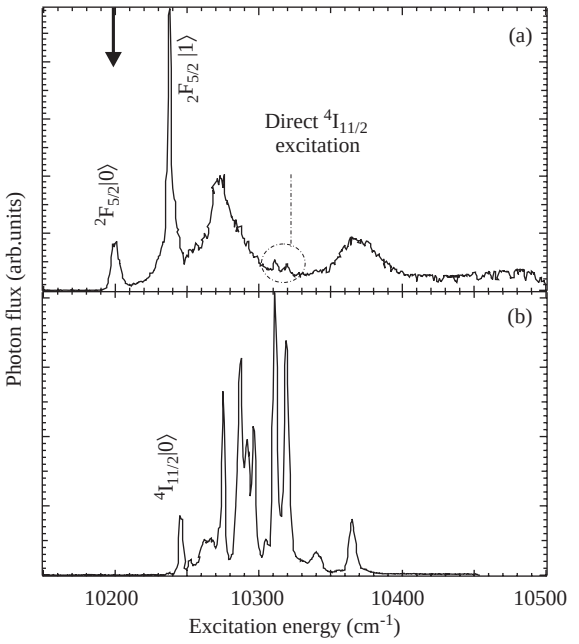


Fig. 2. Excitation spectrum of the 18 200 cm^{-1} emission in (a) $NaYF_4$: 2% Er^{3+}, 18% Yb^{3+} and (b) $NaYF_4$:10% Er^{3+}. The spectra were recorded at (a) 5 K and (b) 12 K. The vertical scales in Figs. 2(a) and (b) are not comparable. The arrow indicates the excitation energy used for obtaining the data shown in Fig. 3.

implies that

$$R_i \equiv p_i \Phi_i \Big/ \sum_{\forall j} p_j \Phi_j \tag{1}$$

will be the fraction of absorbed infra-red photons emitted in band i. The values for R_i at 300 K are shown in Table 1 for different excitation powers.

Note that one important assumption has been made in Eq. (1): it is assumed that intra-excited state emissions do not present a significant contribution to the total emission spectrum. Specifically, it is assumed that the $^4I_{13/2}$ state is mainly populated through a one-photon process rather than through $^4S_{3/2} \rightarrow {}^4I_{13/2}$ emission. Since the photon flux related to this intra-excited state transition represents only a small fraction of the photon flux related to the $^4I_{13/2} \rightarrow {}^4I_{15/2}$ and $^4S_{3/2} \rightarrow {}^4I_{15/2}$ transitions (see the values of the respective f_i as shown in Table 1), this assumption is expected to be valid.

Under this assumption a justified assessment regarding the true efficiency of the $NaYF_4:Er^{3+}$, Yb^{3+} sample is made from the data shown in Table 1: in the high-power limit (i.e. at unfocussed

excitation powers of $\gtrsim 400$ mW) nearly 50% of the NIR photons absorbed are emitted in the three emission bands in the visible ($i = 4, 5, 6$), clearly demonstrating the very high UC efficiency of this material. Furthermore, the data in Table 1 and Fig. 1 also clearly show that the ratio of red (15 000 cm^{-1}) to green (18 200 cm^{-1}) emission is dependent on the excitation power. Thus the output of a device based on $NaYF_4:Er^{3+}$, Yb^{3+} can be tuned to meet specific color-demands when the saturation-limit has not been reached (excitation power $\lesssim 400$ mW).

Fig. 3 shows the temperature dependence of four integrated emission bands when the sample is excited at 10 199 cm^{-1}. Identical results were obtained under 10 238 cm^{-1} excitation. The results for the 11 800 and 15 000 cm^{-1} emission bands are similar to those shown in Fig. 3(c). With the exception of the 6500 cm^{-1} band, as shown in Fig. 3(a), no effect of the excitation power was observed.

Fig. 4 shows the relevant energy levels of Er^{3+} and Yb^{3+} as well as some of the most important energy transfer and light emission processes. Indicated by the sets of two dashed arrows connected with a dotted line are the energy transfer up-conversion (ETU) processes that occur in this material: one Yb^{3+} ion transfers back to the ground state, while simultaneously the Er^{3+} ion transfers to a state of higher energy. This ETU process can occur several times in succession, resulting in an Er^{3+} ion in a high excited state, such as the $^2H_{9/2}$ state at 24 000 cm^{-1} (emission band 6 in Fig. 1), which is reached after three energy transfer steps.

The Yb^{3+} $^2F_{5/2}|0\rangle$ and $^2F_{5/2}|1\rangle$ states are in thermal equilibrium immediately after the excitation. As the $^2F_{5/2}|0\rangle$ state is at too low energy

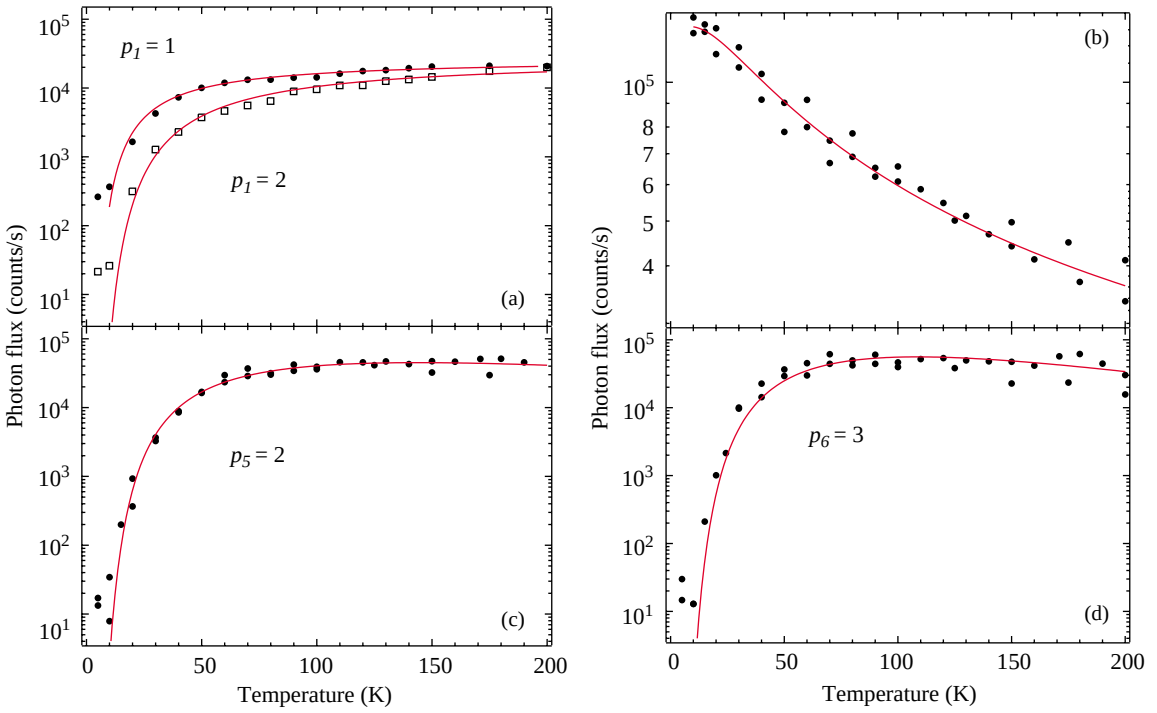


Fig. 3. Temperature dependence of the integrated photon flux of the emission bands centered at (a) 6500, (b) 10 000, (c) 18 200 and (d) 24 000 cm^{-1}, respectively. Excitation was at 10 199 cm^{-1} (indicated by the arrow in Fig. 2(a)) with 100 mW and the vertical scales in Figs. 3(a)–(d) cannot be compared. Note the logarithmic vertical axes. (a) Both the low-power limit ($\sim$ 15 mW: ●) and the high-power limit ($\sim$ 500 mW: □) are shown. The lines through the data are predictions using Eq. (2) for Fig. 3(b) and Eq. (3) for Figs. 3(a), (c), (d) with parameter values $\Delta E = 39\ cm^{-1}$ and p_i as indicated.

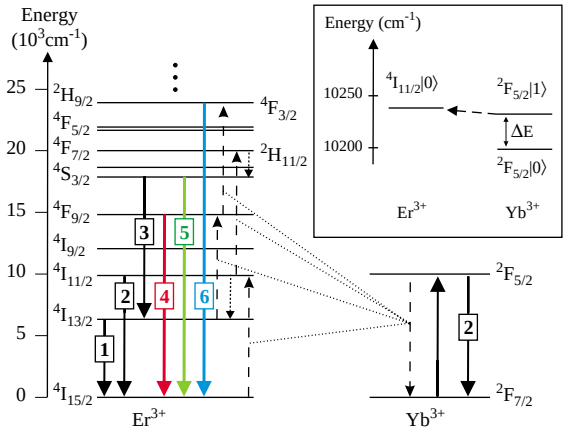


Fig. 4. Schematic energy-level diagram showing luminescent processes (indicated by a solid arrow, the numbers correspond to the six emission bands shown in Fig. 1), Yb^{3+}–Er^{3+} energy-transfer UC processes (set of dashed arrows) and multiphonon relaxation processes on Er^{3+} (dotted arrows). The inset shows the lowest two crystal-field levels within the Yb^{3+} $^2F_{5/2}$ multiplet as well as the lowest level within the Er^{3+} $^4I_{11/2}$ multiplet.

to transfer an excitation to the $^4I_{11/2}|0\rangle$ state (see Fig. 2), this state will luminesce. However, the $^2F_{5/2}|1\rangle$ state will efficiently transfer to the $^4I_{11/2}|0\rangle$ state of Er^{3+}. Based on this, the temperature dependence of the emission band centered at $10\,000\,cm^{-1}$ is determined by the Boltzmann fraction present in the $^2F_{5/2}|0\rangle$ state and the temperature dependence of the Yb^{3+} excitation cross-section, $\sigma_{Yb}(T)$ [14],

$$I_2(T) = I_{2,0}\left(1 - \exp\left[-\frac{\Delta E}{k_B T}\right]\right)\sigma_{Yb}(T), \qquad (2)$$

where ΔE is the energy difference between the $^2F_{5/2}|0\rangle$ and $^2F_{5/2}|1\rangle$ states and $I_{2,0}$ denotes the $10\,000\,cm^{-1}$ intensity at 0 K. A comparable mechanism has recently been suggested to explain the temperature dependence of Tb^{3+} cross-relaxation [15].

Because efficient energy transfer is possible from $^2F_{5/2}|1\rangle$ to $^4I_{11/2}|0\rangle$, the temperature dependence of the other five emission bands will be determined by the Boltzmann fraction present in the $^2F_{5/2}|1\rangle$ state to the power p_i (to account for the number of photons required for excitation of emission band i), multiplied by $[\sigma_{Yb}(T)]^{p_i}$

$$I_i(T) = I_{i,\infty}\exp\left[-\frac{p_i\Delta E}{k_B T}\right][\sigma_{Yb}(T)]^{p_i} \quad (i \neq 2). \qquad (3)$$

Here $I_{i,\infty}$ denotes the intensity of the emission band in the high-temperature limit and ΔE remains the energy difference between the $^2F_{5/2}|0\rangle$ and $^2F_{5/2}|1\rangle$ states. Since Er^{3+} emission from band i requires p_i excitations of Yb^{3+} to begin with, Eq. (3) also depends on $\sigma_{Yb}(T)$ to the power p_i. Note that Eq. (3) implies a direct and (near) resonant energy transfer from the $^2F_{5/2}|1\rangle$ state to the $^4I_{11/2}|0\rangle$, and that no phonon-mediated step is required. As a result, this process is insensitive to the phonon density of states in the $NaYF_4$ host-lattice.

Eqs. (2) and (3) determine the temperature dependence of the six emission bands, since ΔE ($39\,cm^{-1}$, as found from Fig. 2) and the p_i (see Table 1) are known. The lines through the data in Fig. 3(a)–(d) clearly show that the experimental data is predicted very well, with the exception of the high-power data (squares) of the $6500\,cm^{-1}$ emission band shown in Fig. 3(a): its temperature dependence does not correspond to a one-photon process but is predicted well using $p_1 = 2$, as shown by the line through the data. Thus the $^4I_{13/2}$ population becomes predominately a two-photon process at high excitation power. This effect is attributed to the importance of cross-relaxation in the high-power limit. Due to the large $^4S_{3/2}$ population, the $|^4S_{3/2}, ^4I_{15/2}\rangle \Rightarrow |^4I_{9/2}, ^4I_{13/2}\rangle$ cross-relaxation process becomes important. As a result, two NIR excitation photons are used to populate the $^4I_{13/2}$ state in the high-power limit, while in the low-power limit the state is populated through a non-radiative $^4I_{11/2} \rightarrow {}^4I_{13/2}$ multiphonon relaxation. Furthermore, this limits the applicability of Eq. (1), suggesting that the high-power values for R_i (shown in Table 1) may still be somewhat underestimated. These two competing mechanisms for populating the $^4I_{13/2}$ state have further consequences in the excitation-power dependence of the ratio of red to green emission (as shown in Fig. 2), since $^4F_{9/2}$ is populated from $^4I_{13/2}$ through ETU. As a result, the $^4F_{9/2}$ population, and thus the red emission, will become a three-photon process at high

excitation power, changing the red to green emission ratio.

4. Conclusions

In conclusion, this study provides quantitative data on the efficiency of near-infrared to visible UC in $NaYF_4:Er^{3+},Yb^{3+}$ and demonstrates that the material is very efficient: in the high-power limit over 30% of all photons are emitted in the visible, which corresponds to $\sim 50\%$ of all near-infrared photons absorbed. The huge temperature dependence observed for all emission bands is explained through efficient energy transfer from the second crystal field level of the $^2F_{5/2}$ multiplet in Yb^{3+} to the lowest level of the $^4I_{11/2}$ multiplet in Er^{3+} and efficient UC on Er^{3+}. The $6500\,cm^{-1}$ emission from Er^{3+} has a temperature dependence that changes with excitation power: at high excitation power, this level is significantly populated through a two-photon cross-relaxation process.

Acknowledgements

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